# $\mathrm{EFX}_0$ Allocations with at Most Three Relevant Goods

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**Abstract.** An allocation of indivisible goods is $\mathrm{EFX}_0$ if no agent envies another agent's bundle after removing any single good from it, even one the envious agent values at zero. We prove that every instance with nonnegative additive valuations in which each agent positively values at most three goods has a complete $\mathrm{EFX}_0$ allocation in which all bundles but at most one have at most two goods, and an algorithm, Draft and Exchange (DE), that computes one in time $O(n(n+m))$ with at most $4n$ trades. After a draft, goods are traded along a chain or a ring of agents, each of which gets strictly better off, until a free agent can take all the leftover goods (or every agent holds its pair). The key fact is that when no free agent can take them, every agent can point at an agent that would gladly take its good, and the pointers close a ring. The proof is machine-checked in Lean, together with the algorithm, its bound of $4n$ trades and its running time.

**Keywords:** Fair division · Indivisible goods · EFX · Trading cycles

## 1 Introduction

Agents $N$ ($|N| = n \geq 1$) share a set $M$ of $m$ indivisible goods with nonnegative additive valuations $v_i(S) = \sum_{g \in S} v_i(g)$. An allocation $X$ assigns every good to exactly one agent, who receives the bundle $X_i$; no good is left out (allocations are *complete*). It is *EFX*$_0$ if

$$v_i(X_i) \;\geq\; v_i(X_j \setminus \{h\}) \qquad \text{for all } i \neq j \text{ and all } h \in X_j,$$

and *EFX* asks this only for goods $h$ with $v_i(h) > 0$ [3]. So $\mathrm{EFX}_0$ implies EFX, and the two coincide for positive values. Under $\mathrm{EFX}_0$, if $X_j$ contains a good that $i$ values at 0, then $i$ must not envy $X_j$ at all. Whether complete EFX allocations always exist for additive valuations is open; they exist in special cases [5,7,12,2,4,1,6,13,8,9] (Table 1 in Appendix F). Let $R_i = \{g : v_i(g) > 0\}$, the *relevant* goods of $i$. If $|R_i| \leq 2$ for all $i$, serial dictatorship gives $\mathrm{EFX}_0$ [11]. At three goods this breaks: an agent that values $a$, $b$, $c$ at 4, 3, 2 may take $a$ and then see $b$ and $c$ end up in the last agent's bundle with a third good; removing that third good leaves $b$ and $c$, worth $5 > 4$.

**Theorem 1 (existence and shape).** *Every instance with $n \geq 1$ agents and nonnegative additive valuations in which every agent positively values at most three goods has a complete $EFX_0$ allocation in which all bundles but at most one have at most two goods.*

**Theorem 2 (algorithm).** *Algorithm* DE *(Section 6) computes such an allocation with at most $n$ peeling steps and at most $4n$ trades, in time $O(n(n+m))$. On real values, every decision of* DE *compares two sums of at most two values of one agent.*

*What is new.* As far as we know,[1] no earlier result settles the case of three goods. Viswanathan and Mehta [14] study the same restriction, up to four goods per agent, but for ordinary EFX and with only a proof sketch; EFX does not give $\mathrm{EFX}_0$, and the difference matters exactly here, where most goods are worth 0 to most agents (their algorithm gives the goods nobody values to an arbitrary agent, which can break $\mathrm{EFX}_0$). Every other existence result for complete $\mathrm{EFX}_0$ allocations restricts something our model leaves free: the number of agents, of goods compared with agents, of agents valuing a good, the hypergraph girth, or the number of distinct valuations. For example, six agents valuing $\{g_0, g_1, g_6\}$, $\{g_0, g_1, g_7\}$, $\{g_0, g_2, g_8\}$, $\{g_2, g_3, g_4\}$, $\{g_3, g_5, g_9\}$, $\{g_4, g_5, g_{10}\}$ with distinct values in $(1, 2)$ are covered by none of them ($m = n + 5$, three agents value $g_0$, the hypergraph has girth 2, six valuations). We contribute the existence (Theorem 1); the shape, which cannot be improved in the obvious ways (Appendix D) and has no counterpart we know of; the algorithm (Theorem 2); and a complete machine-checked proof.

*How the proof works.* Agents that can leave with at most one good do so; the others draft at most one good each. Then, until a *free* agent can take all the leftover goods (or every agent holds its pair), goods are traded along a *chain* or a *ring* of agents, each of which gets strictly better off. Section 6 ends with a small example.

## 2 Peeling

The proofs of Sections 2 and 3 are in Appendix A.

Agents and goods are numbered, and "index order" and "the first" refer to these numbers. A good is *alone* in an allocation if it is the only good of its bundle.

[1] Literature search of 25 September 2026 (arXiv, the web, Zenodo, Google Scholar, Crossref, the EC 2026 programme), recorded in `proofs/novelty.md` of [10].

**Lemma 1 (peeling).** *Let $i \in N$ and $P \subseteq M$ with (i) $v_i(P) \geq v_i(M \setminus P)$ and (ii) $|P| \leq 1$. If the instance with agents $N \setminus \{i\}$ and goods $M \setminus P$ has an EFX$_0$ allocation $Z$, then $Z$ together with $X_i = P$ is an EFX$_0$ allocation of the whole instance.*

*Rule R1.* Let $G$ be the set of remaining goods, and $p$ agent $i$'s favourite good of $G$ (largest value; between equal values, the lower index). If $G = \emptyset$ or $v_i(p) = 0$, so that $i$ values no remaining good, then $i$ may leave with $P = \emptyset$. If $v_i(p) > 0$ and $v_i(G \setminus \{p\}) \leq v_i(p)$, then $i$ may leave with $P = \{p\}$. Rule R1 applies, for instance, to every agent that values at most two remaining goods; so if every agent values at most two goods, serial dictatorship is EFX$_0$ [11].
When $|R_i| = 3$ we name agent $i$'s relevant goods $a_i$, $b_i$, $c_i$ so that $v_i(a_i) \geq v_i(b_i) \geq v_i(c_i) > 0$, breaking ties in favour of the lower index. This gives a strict *ranking* $a_i \succ_i b_i \succ_i c_i$, and $g \succ_i h$ implies $v_i(g) \geq v_i(h)$. Agent $i$ is *strictly balanced* if $v_i(a_i) < v_i(b_i) + v_i(c_i)$.

**Lemma 2 (when nobody can leave).** *Let agent $i$ value at most three goods of the set $G$ of remaining goods, and suppose that rule R1 does not apply to $i$. Then $i$ values exactly three goods of $G$ and is strictly balanced with respect to them.*

In Sections 3–5 we are in the *core case*: every agent values exactly three goods and is strictly balanced. Goods that no agent values may exist.

## 3 States

**Definition 1 (states and scores).** *A* state *$Y$ gives each agent $i$ a* holding *$Y_i$: nothing, $\{a_i\}$, $\{b_i\}$, $\{c_i\}$, or its* pair *$\{b_i, c_i\}$, with the holdings pairwise disjoint. $U$ is the set of agents holding their pair. The goods held by nobody are* leftover*; $J$ is the set of leftover goods. Agent $i$'s* score *is 4, 3, 2, 1 or 0 if it holds its pair, $a_i$, $b_i$, $c_i$ or nothing. Since $v_i(a_i) < v_i(b_i)+v_i(c_i)$, a higher score never means a lower value.*

**Definition 2 (wants, validity, free agents).** *Let $Y$ be a state. An agent $i \notin U$* wants *a good $g \in R_i$ if it holds nothing, or holds a good $y$ with $g \succ_i y$. Agents in $U$ want nothing. $Y$ is* valid *if every good that some agent wants is held alone: it is the whole holding of some agent. An agent $i \notin U$ is* free *if it holds nothing or holds a good that nobody wants. $F$ is the set of free agents.*

No agent wants its own good, so in a valid state the holder of a wanted good is another agent, outside $U$ and not free. Leftover goods and the goods of pair holders are not wanted in a valid state. An agent outside $U \cup F$ holds a single good that some agent wants.

**Lemma 3 (the draft).** *Let the agents, in any order, each take the first of $a_i$, $b_i$, $c_i$ not yet taken, or nothing if all three are taken. The result is a valid state with $U = \emptyset$.*

**Lemma 4 (staying valid).** *Let $Y$ be valid, and $Y'$ a state in which every agent's score is at least its score in $Y$ and every good wanted in $Y$ is held alone. Then $Y'$ is valid.*

## 4 Finishing

**Definition 3 (blockers, finishing).** *Let $Y$ be a state and $o$ a free agent. A* blocker *of $o$ is an agent $x \neq o$ that holds exactly its top, $Y_x = \{a_x\}$, and has $b_x, c_x \in J \cup Y_o$. Agent $o$* can finish *with a set $H \subseteq J$ if $|H| \leq |F| - 1$ and $H$ contains $b_x$ or $c_x$ for every blocker $x$ of $o$. Its* completion *$X$ gives each good of $H$ to a different free agent other than $o$, gives $o$ the bundle $X_o = Y_o \cup (J \setminus H)$, and lets every other agent keep its holding (plus its good of $H$, if any).*

So a blocker of $o$ is an agent that would find both $b_x$ and $c_x$ in $o$'s bundle if $o$ took all the leftover goods; $H$ moves one of the two out of it. When every agent holds its pair, the completion with any agent $o$ gives $o$ the bundle $Y_o \cup J$ and lets the others keep their pairs.

**Theorem 3 (soundness).** *Let $Y$ be a valid state, and $X$ the completion with a free agent $o$ that can finish with a set $H$, or, if every agent holds its pair, with any agent $o$. Then $X$ is an EFX$_0$ allocation, and every bundle other than $X_o$ has at most two goods.*

*Proof. Sizes.* A free agent other than $o$ holds at most one good and receives at most one good of $H$; a pair holder holds two goods; every other agent keeps a single good.

*Wanted goods stay alone.* Let $g$ be wanted, held alone by $j$. Then $j$ is not free, so $j \neq o$ and $j$ receives no good of $H$: $X_j = \{g\}$. Hence every good in a bundle of two or more goods is unwanted.

*No envy.* Let $i \neq j$ and $h \in X_j$. We show $v_i(X_j \setminus \{h\}) \leq v_i(X_i)$, where $v_i(X_i) \geq v_i(Y_i)$. If $|X_j| \leq 1$, then $X_j \setminus \{h\} = \emptyset$. Let $|X_j| \geq 2$, and $T = X_j \cap R_i$; it contains no good that $i$ wants and none of $i$'s holding.

- If $i$ holds its pair, then $T \subseteq \{a_i\}$, and $v_i(T) \leq v_i(a_i) < v_i(b_i) + v_i(c_i)$.
- If $i$ holds nothing, it wants all of $R_i$, so $T = \emptyset$.
- If $Y_i = \{y\}$, every good of $T$ is ranked below $y$ and worth at most $v_i(y)$. If $|T| \leq 1$, then $v_i(X_j \setminus \{h\}) \leq v_i(T) \leq v_i(y)$. If $|T| = 2$, then $y = a_i$ and $T = \{b_i, c_i\}$. If $X_j = \{b_i, c_i\}$, removing $h$ leaves one good, worth at most $v_i(b_i) \leq v_i(a_i)$. Otherwise $|X_j| \geq 3$, so $j = o$ by the sizes, and $b_i, c_i \in X_o \subseteq Y_o \cup J$: $i$ is a blocker of $o$. Then $H$ contains $b_i$ or $c_i$, which went to an agent other than $o$, a contradiction. (When every agent holds its pair, only the first case occurs.) □

## 5 Improving

**Definition 4 (rings).** *Let $Y$ be a state. A* ring *is a cycle $w_0 \to w_1 \to \cdots \to w_{k-1} \to w_0$ of $k \geq 2$ distinct agents outside $U$, each holding a single good $y_t = y_{w_t}$, such that each arrow $w_t \to w_{t+1}$ (indices modulo $k$) is*

- *a* want arrow*: $w_{t+1}$ wants $y_t$; or*
- *a* pair arrow*: $w_t$ is free, $w_{t+1}$ is a blocker of $w_t$, and the pair of $w_{t+1}$ is $\{y_t, h_{t+1}\}$ with $h_{t+1} \in J$;*

*and the leftover goods $h_{t+1}$ of the pair arrows are pairwise distinct.* Trading along the ring *gives each $w_{t+1}$ the good $y_t$, together with $h_{t+1}$ for a pair arrow, in place of its own good; agents off the ring keep their holdings.*

An arrow from a free agent is a pair arrow (nobody wants its good), and an arrow from any other agent is a want arrow.

**Lemma 5 (ring).** *Trading along a ring of a valid state gives a valid state in which every agent of the ring has a higher score and every other agent the same score.*

*Proof. A state.* Each good $y_t$ moves to $w_{t+1}$ only, and the goods $h_{t+1}$ are distinct leftover goods; agents off the ring keep goods that are neither. The new holdings are a single good of the receiver or its pair. *Scores.* Along a want arrow, $w_{t+1}$ receives a good it ranks above what it held. Along a pair arrow, $w_{t+1}$ goes from its top to its pair. *Wanted goods.* Let $g$ be wanted in $Y$, held alone by $j$. If $j$ is off the ring, it keeps $g$. If $j = w_t$, then $w_t$ is not free, so its arrow is a want arrow, and $w_{t+1}$ now holds $g$ alone. Lemma 4 gives validity. □

**Lemma 6 (chain).** *Let $Y$ be valid, and $x$ an agent with $Y_x = \{a_x\}$ and $b_x, c_x \in J$. Let $j_0 = x$, and as long as $j_t$ holds a good that some agent wants, let $j_{t+1}$ be the first such agent. If an agent repeats, the agents from its first occurrence on form a ring of want arrows. Otherwise the sequence stops at an agent $j_k$ ($k \geq 0$) that holds nothing or a good nobody wants, and the following gives a valid state in which $x, j_1, \ldots, j_k$ have higher scores and every other agent the same score: $x$ takes its pair; each $j_t$ with $1 \leq t \leq k$ takes the good of $j_{t-1}$; the old good of $j_k$, if any, becomes leftover.*

*Proof.* Agent $x$ holds its top and wants nothing, so $j_t \neq x$ for $t \geq 1$. If $j_s = j_t$ with $1 \leq s < t$ is the first repeat, then $j_s, \ldots, j_{t-1}$ are distinct agents outside $U$, each holding a good that the next one wants: a ring of want arrows ($t - s \geq 2$, as no agent wants its own good). Otherwise: the goods $b_x$, $c_x$ were leftover, and each moved good moves one step, so this is a state. Agent $x$ goes from its top to its pair, and each $j_t$, $t \geq 1$, receives a good it wants. A wanted good is not $b_x$ or $c_x$ (they are leftover); if its holder is off the path it stays, and if its holder is $j_t$, then $t < k$ (the good of $j_k$ is not wanted), and $j_{t+1}$ now holds it alone. Lemma 4 gives validity. □

Applying a chain, or trading along a ring, is a *trade*. From now on, assume that no chain applies:

**(NC)** no agent $x$ with $Y_x = \{a_x\}$ has both $b_x$ and $c_x$ in $J$.

**Lemma 7 (the finishing test).** *Let $Y$ be a state with (NC), and $o$ a free agent.*

*(a) If $o$ holds nothing, it has no blocker.*
*(b) If $x$ is a blocker of $o$, then $o$ holds a good $y_o$, and the pair of $x$ is $\{y_o, h_x\}$ for a leftover good $h_x$.*
*(c) Let $H_o$ be the set of the goods $h_x$ of the blockers $x$ of $o$. Then $o$ can finish if and only if $|H_o| \leq |F| - 1$, and then it can finish with $H_o$. In particular a free agent holding nothing can finish.*

*Proof.* Let $x$ be a blocker of $o$: $b_x, c_x \in J \cup Y_o$, and not both are in $J$, by (NC). So one of them is in $Y_o$, which proves (a), and, as $o$ holds at most one good, that good is $y_o$ and the other one, $h_x$, is leftover: (b). A set $H \subseteq J$ contains $b_x$ or $c_x$ if and only if it contains $h_x$; so the sets $H$ allowed by Definition 3 are those with $H_o \subseteq H \subseteq J$ and $|H| \leq |F| - 1$, which exist if and only if $|H_o| \leq |F| - 1$. If $o$ holds nothing, $H_o = \emptyset$ and $|F| \geq 1$. □

**Theorem 4 (Improvement Lemma).** *Let $Y$ be a valid state in which not every agent holds its pair and no free agent can finish. Then a chain or a ring gives a valid state in which every agent has at least its score and some agent a higher one.*

*Proof.* If (NC) fails, Lemma 6 applies. Assume (NC). By Lemma 7, every free agent $o$ holds a good and has $|H_o| \geq |F|$.

*Distinct leftover goods.* List the free agents in index order, $o_1, \ldots, o_f$. For $i = 1, \ldots, f$ in turn, let $x_{o_i}$ be the first blocker of $o_i$ whose leftover good is not among $q_{o_1}, \ldots, q_{o_{i-1}}$, and $q_{o_i} = h_{x_{o_i}}$. It exists because the blockers of $o_i$ have $|H_{o_i}| \geq f > i - 1$ different leftover goods.

*Pointers.* For every agent $j \notin U$ let $\sigma(j) = x_j$ if $j$ is free, and otherwise let $\sigma(j)$ be the first agent that wants the good of $j$ (one exists, as $j \notin U \cup F$). In both cases $\sigma(j) \notin U$ and $\sigma(j) \neq j$.

*The ring.* Start at the first agent outside $U$ and follow $\sigma$ until an agent repeats. The agents from its first occurrence on form a cycle $w_0 \to \cdots \to w_{k-1} \to w_0$ with $k \geq 2$, each holding a single good (free agents by Lemma 7). The arrow from a free agent $w_t$ is a pair arrow, with $h_{t+1} = q_{w_t}$, by Lemma 7(b); these goods are distinct for distinct free agents. Every other arrow is a want arrow. So the cycle is a ring, and Lemma 5 applies. □

## 6 The Algorithm Draft and Exchange

Algorithm 1 lists DE; ties are broken by index as in Section 2.

**Theorem 5 (DE is correct).** *On every instance in which every agent values at most three goods,* DE *returns a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods. It peels at most $n$ agents and makes at most $4n$ trades.*

*Proof. Peeling.* Each round removes one agent, so there are at most $n$ rounds. Rule R1 for the agent $i$ peeled with $P_i$ is conditions (i) and (ii) of Lemma 1 on the instance left before it, so by induction from the last round down it suffices that DE returns an EFX$_0$ allocation of the agents $A$ and goods $G$ left at the end. If $|A| = 1$, the single agent faces no other bundle.

*The core.* Let $|A| \geq 2$. Rule R1 applies to no agent of $A$, so by Lemma 2 this is the core case. The draft gives a valid state (Lemma 3). A chain or a ring gives a valid state in which nobody's score drops and some score rises (Lemmas 6 and 5; the ring exists when it is used, by the proof of

Theorem 4). The total score is an integer between 0 and $4|A|$, so there are at most $4|A| \le 4n$ trades, and the loop stops. When it stops, $o$ can finish with $H$ (Lemma 7(c); the chain test failed, so (NC) holds), or every agent holds its pair; Theorem 3 gives an $\mathrm{EFX}_0$ allocation in which only $X_o$ can have more than two goods. Peeled agents receive at most one good. □

*A small example.* Agents $z$, $x$, $o$, in index order, rank $g_4 \succ g_0 \succ g_1$, $g_0 \succ g_1 \succ g_2$ and $g_0 \succ g_4 \succ g_1$, with values 4, 3, 2; nobody values $g_3$. Nobody is peeled. In the draft $z$ takes $g_4$, $x$ takes $g_0$, and $o$ takes $g_1$; the leftover goods are $g_2$, $g_3$. Agent $o$ wants $g_0$ and $g_4$, which are held alone: the state is valid, and $o$ is the only free agent. No chain applies. Agent $x$ is a blocker of $o$: if $o$ took $\{g_1, g_2, g_3\}$, $x$ would remove $g_3$ and see $g_1$ and $g_2$, worth $5 > 4$. So $H_o = \{g_2\}$, there is no other free agent to take $g_2$, and $o$ cannot finish. The pointers are $\sigma(o) = x$, a pair arrow ($x$'s pair is $o$'s good $g_1$ and the leftover $g_2$), and $\sigma(x) = \sigma(z) = o$, want arrows ($o$ wants $g_0$ and $g_4$). Following them from $z$ closes the ring $o \to x \to o$: $x$ takes $\{g_1, g_2\}$ and $o$ takes $g_0$. Now nobody wants anything, the free agents are $z$ and $o$, and $z$ has no blocker, so $z$ finishes: $X_z = \{g_4, g_3\}$, $X_x = \{g_1, g_2\}$, $X_o = \{g_0\}$, an $\mathrm{EFX}_0$ allocation.

**Algorithm 1** Draft and Exchange (DE)

**Require:** agents $N$, goods $M$, nonnegative values $v_i(g)$, every agent with $|R_i| \le 3$

$A \leftarrow N$, $G \leftarrow M$
**while** $|A| \ge 2$ and rule R1 applies to an agent of $A$, for $G$ **do** ▷ peeling
  $i \leftarrow$ the first such agent; $X_i \leftarrow P_i$ (by rule R1); $A \leftarrow A \setminus \{i\}$, $G \leftarrow G \setminus P_i$
**if** $|A| = 1$ **then** give $G$ to the agent of $A$ and **return** $X$
**for** $i \in A$ in index order **do** ▷ the draft
  $Y_i \leftarrow$ the first of $\{a_i\}, \{b_i\}, \{c_i\}$ whose good is not taken, else nothing
**loop**
  **if** some $x$ with $Y_x = \{a_x\}$ has $b_x, c_x \in J$ **then** ▷ chain
    apply the chain of the first such $x$, or the ring of want arrows it runs into (Lemma 6)
  **else if** every agent holds its pair **then** ▷ finish
    $o \leftarrow$ the first agent; $H \leftarrow \emptyset$; **break**
  **else if** some free agent $o$ has $|H_o| \le |F| - 1$ **then** ▷ finish
    $o \leftarrow$ the first such agent; $H \leftarrow H_o$; **break**
  **else** ▷ ring
    trade along the ring of the proof of Theorem 4
give the $k$-th good of $H$ (in the order of first blockers) to the $k$-th free agent other than $o$
$X_o \leftarrow Y_o \cup (J \setminus H)$; every other $i \in A$ keeps $Y_i$ (plus its good of $H$, if any)
**return** $X$

*Running time and real values.* With an array of the holder of each good, a loop iteration or a peeling round takes $O(n + m)$ steps, so DE takes $O(n(n+m))$, reading the input included (Appendix E). Every decision of DE compares a value of an agent with another or with the sum of two of its values, and the proofs use nothing else about the values. So Theorem 5 and these remarks prove Theorems 1 and 2, also for real values.

## 7 Further Results, Verification and Open Problems

Appendix B has a larger example, where DE's ring gives pairs to two agents at once; rings may need any number of pair arrows (Appendix C), and some instances need a bundle of three or four goods (Appendix D). Every result here is machine-checked in core Lean 4 (`lean/EFX/K3DE*.lean` in [10]); tests found no failure; the reviews were by AI agents, with no human peer review yet (Appendix E). Open: four relevant goods per agent [14]; a better bound than $4n$ trades; the least size of the large bundle.

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## A Proofs of Sections 2 and 3, and a Corollary

*Proof (of Lemma 1).* Agent $i$ and a bundle $Z_j$: $v_i(Z_j \setminus \{h\}) \le v_i(Z_j) \le v_i(M \setminus P) \le v_i(P)$. Agent $j \neq i$ and the bundle $P$: removing its only good leaves nothing. Agent $j \neq i$ and a bundle $Z_k$: this is a condition of $Z$. □

*Proof (of Lemma 2).* Rule R1 fails, so $v_i(p) > 0$ and $v_i(G \setminus \{p\}) > v_i(p)$. If $i$ valued at most two goods of $G$, then $v_i(G \setminus \{p\})$ would be the value of at most one good, at most $v_i(p)$. So $i$ values exactly three goods of $G$, $p$ is its top, and $v_i(b_i) + v_i(c_i) > v_i(a_i)$. □

*Proof (of Lemma 3).* Each agent holds one of its goods or nothing, and no good is taken twice. If $i$ wants $g$, then $g$ was taken before $i$'s turn, by an earlier agent as its only good, and holdings never change during the draft. □

*Proof (of Lemma 4).* Let $i$ want $g$ in $Y'$. If $i$ holds nothing in $Y'$, its score in $Y$ was 0, so it held nothing in $Y$ and wanted $g$ there too. If $i$ holds a single good $y'$ in $Y'$, its score in $Y$ was below 4, so in $Y$ it held nothing, or a good $y$ with $y' = y$ or $y' \succ_i y$; since $g \succ_i y'$, again $i$ wanted $g$ in $Y$. So $g$ is held alone in $Y'$. □

**Corollary 1 (Pareto-optimal states).** *In the core case, in every valid state that no valid state Pareto-dominates in scores, either every agent holds its pair or some free agent can finish. In particular every core instance has a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods.*

*Proof.* The first part is Theorem 4. Valid states exist (Lemma 3), there are finitely many, and one with the largest total score is Pareto-optimal; Theorem 3 finishes. □

## B A Worked Example

On this instance every valid state that Pareto-dominates the draft state gives pairs to two agents at once, so a ring with a single pair arrow is not enough (Appendix C). Ten goods $g_0, \ldots, g_9$ and six agents, in index order $x_1$, $x_1'$, $x_2$, $x_2'$, $o_1$, $o_2$, with values 4, 3, 2 on their goods $a \succ b \succ c$ (any strictly balanced values with these rankings give the same run):

| agent | $a \succ b \succ c$ | after the draft | after the trade |
|---|---|---|---|
| $x_1$ | $g_0 \succ g_4 \succ g_6$ | $\{g_0\}$ $(a)$ | $\{g_4, g_6\}$ (pair) |
| $x_1'$ | $g_1 \succ g_4 \succ g_7$ | $\{g_1\}$ $(a)$ | $\{g_1\}$ $(a)$ |
| $x_2$ | $g_2 \succ g_5 \succ g_8$ | $\{g_2\}$ $(a)$ | $\{g_5, g_8\}$ (pair) |
| $x_2'$ | $g_3 \succ g_5 \succ g_9$ | $\{g_3\}$ $(a)$ | $\{g_3\}$ $(a)$ |
| $o_1$ | $g_2 \succ g_3 \succ g_4$ | $\{g_4\}$ $(c)$ | $\{g_2\}$ $(a)$ |
| $o_2$ | $g_0 \succ g_1 \succ g_5$ | $\{g_5\}$ $(c)$ | $\{g_0\}$ $(a)$ |

*The draft.* Nobody is peeled. The four $x$'s take their tops; $o_1$ finds $g_2$ and $g_3$ taken and takes $g_4$; $o_2$ takes $g_5$. Agent $o_1$ wants $g_2$ and $g_3$, $o_2$ wants $g_0$ and $g_1$, and the $x$'s want nothing; the four wanted goods are held alone. The leftover goods are $g_6, \ldots, g_9$, and the free agents are $o_1$ and $o_2$. No chain applies: the $b$ of each $x$ is held by $o_1$ or $o_2$.

*Nobody can finish.* The blockers of $o_1$ are $x_1$ (pair $\{g_4, g_6\}$) and $x_1'$ (pair $\{g_4, g_7\}$), so $H_{o_1} = \{g_6, g_7\}$; symmetrically $H_{o_2} = \{g_8, g_9\}$. Both have $|H_o| = 2 > |F| - 1$. Concretely, if $o_1$ takes the leftovers, only $o_2$ can

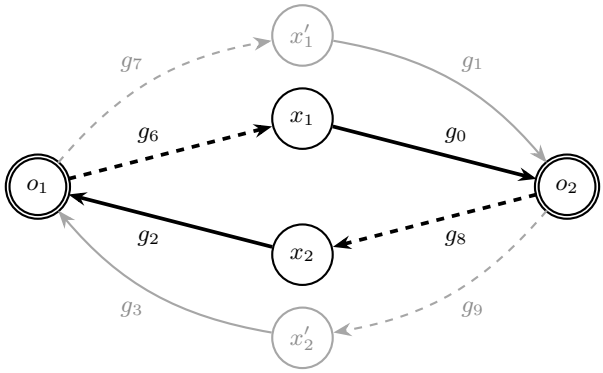



**Fig. 1.** The arrows of the draft state of Appendix B. Double circles are the free agents. Dashed arrows are pair arrows, labelled by the blocker's leftover good; solid arrows are want arrows, labelled by the good that moves. The ring chosen by DE is drawn in black. Every cycle of these arrows has four agents and two pair arrows.

take one of $g_6$, $g_7$; with $H = \{g_6\}$, $X_{o_1} = \{g_4, g_7, g_8, g_9\}$, and $x'_1$, holding $g_1$ (worth 4), values $X_{o_1}$ without $g_8$ at $3 + 2 = 5$. All ten candidate completions (finisher $o_1$ or $o_2$, with $H$ empty or one leftover good) fail the $\text{EFX}_0$ test.

*No small trade helps.* Every want arrow ends at $o_1$ or $o_2$, so a ring with no pair arrow would have to leave one of them by a want arrow; but nobody wants their goods. A ring with exactly one pair arrow, from $o_1$ say, would return to $o_1$ by want arrows; but $x_1$ and $x'_1$ have want arrows only to $o_2$, and the arrows from $o_2$ are pair arrows. In fact exactly four valid states Pareto-dominate the draft state, and each gives pairs to two agents.

*The trade.* DE takes the first blocker of each free agent: $x_{o_1} = x_1$, with $q_{o_1} = g_6$, and $x_{o_2} = x_2$, with $q_{o_2} = g_8$. So $\sigma(o_1) = x_1$, $\sigma(x_1) = o_2$ ($o_2$ wants $g_0$), $\sigma(o_2) = x_2$ and $\sigma(x_2) = o_1$ ($o_1$ wants $g_2$): the ring of Figure 1. Along it, $x_1$ takes $\{g_4, g_6\}$, $o_2$ takes $g_0$, $x_2$ takes $\{g_5, g_8\}$ and $o_1$ takes $g_2$. The total score rises from 14 to 20.

*Finishing.* Now nobody wants anything, the leftover goods are $g_7$ and $g_9$, and the free agents are $x'_1$, $x'_2$, $o_1$, $o_2$. No chain applies ($b$ of $x'_1$ is $g_4$, now held by $x_1$; similarly for the others). The first free agent, $x'_1$, has no blocker: no other agent holding its top has both $b$ and $c$ in $\{g_1, g_7, g_9\}$. So

$x'_1$ finishes with $H = \emptyset$, and DE returns

$$X_{x_1} = \{g_4, g_6\}, \quad X_{x'_1} = \{g_1, g_7, g_9\}, \quad X_{x_2} = \{g_5, g_8\},$$
$$X_{x'_2} = \{g_3\}, \quad X_{o_1} = \{g_2\}, \quad X_{o_2} = \{g_0\}.$$

Directly: the only bundle of three goods, $X_{x'_1}$, is worth 3 to $o_2$ (only $g_1$), which holds $g_0$, worth 4, and 2 to $x'_2$ (only $g_9$), which holds $g_3$, worth 4; after removing one good, a bundle of two goods is worth at most 3 to any other agent, and every agent holds at least 4.

## C How Large Must a Ring Be?

Call a *short move* a chain or a ring with at most one pair arrow. A natural simpler algorithm would use only short moves. The next proposition shows that the instance of Appendix B is as small as possible in its number of agents among states where short moves are not enough.

**Proposition 1 (when short moves are not enough).** *In the core case, let $Y$ be a valid state in which not every agent holds its pair, no free agent can finish, and no short move applies. Then $|F| \geq 2$, $n \geq 6$ and $m \geq 8$; if $|F| \geq 3$, then $n \geq 9$ and $m \geq 12$. The bounds $n = 6$ and $m = 8$ are attained.*

*Proof.* No chain applies, so (NC) holds, and by Lemma 7 every free agent $o$ holds a good and has $|H_o| \geq |F|$, hence at least $|F|$ blockers. The blockers of two different free agents $o, o'$ are different: a common blocker would have the pair $\{y_o, y_{o'}\}$, with no leftover good, against Lemma 7(b). There is no ring of want arrows, so a walk that repeatedly moves to an agent that wants the current agent's good never repeats an agent; it ends at an agent whose good nobody wants, or that holds nothing, which is free. Every agent holds a good: pair holders two, free agents one by Lemma 7, others a wanted good. Let $k = |F|$; $k \geq 1$, because the walk from an agent outside $U$ ends at a free agent.

If $k = 1$, let $F = \{o\}$ and $x$ a blocker of $o$. Then $x$ is not free, and its walk ends at $o$; with the pair arrow $o \to x$ this is a ring with one pair arrow, a short move. So $k \geq 2$.

If $k = 2$, let $F = \{o_1, o_2\}$. Agent $o_1$ has at least two blockers and only $o_2$ is another free agent, so some blocker $x$ of $o_1$ is not free. Its walk cannot end at $o_1$, as above, so it ends at $o_2$, entering it by a want arrow ($x \neq o_2$). So $o_2$ wants a good and does not hold its top: $o_2$ is not a blocker of $o_1$,

and no blocker of $o_1$ is free. The same holds for $o_2$. So $n \geq 2+2+2 = 6$, and $m \geq n + |J| \geq 6 + 2 = 8$.

If $k \geq 3$, the blockers of the $k$ free agents are at least $k^2$ different agents, at most $k$ of them free, so $n \geq |F| + (k^2 - k) = k^2 \geq 9$, and $m \geq n + |J| \geq k^2 + k \geq 12$.

*Attained.* The instance of Appendix B has $n = 6$ ($m = 10$). For $m = 8$: $o_1$ ranks $g_4 \succ g_5 \succ g_0$, $o_2$ ranks $g_6 \succ g_7 \succ g_1$, $x_1$ ranks $g_4 \succ g_1 \succ g_2$, $x_1'$ ranks $g_5 \succ g_3 \succ g_1$, $x_2$ ranks $g_6 \succ g_0 \succ g_2$, $x_2'$ ranks $g_7 \succ g_3 \succ g_0$, with $o_1$ holding $g_0$, $o_2$ holding $g_1$ and every $x$ its top. Then $J = \{g_2, g_3\}$, $F = \{o_1, o_2\}$, the blockers of $o_2$ are $x_1$, $x_1'$ (leftover goods $g_2$, $g_3$) and those of $o_1$ are $x_2$, $x_2'$ (leftover goods $g_2$, $g_3$); nobody can finish, and no short move applies. □

In the second instance the distinct leftover goods matter: of the four cycles $o_2 \to x \to o_1 \to x' \to o_2$ with two pair arrows, two use the same leftover good twice, and trading along them would give it to two agents; the other two are rings.

*No bounded number of pair arrows suffices.* For every $k \geq 2$ and $d \geq 1$ there are valid states with $k$ free agents $o_0, \ldots, o_{k-1}$ in which each $o_i$ is the root of a binary tree of depth $d$ whose agents want the goods of their two children, and the leaves of the tree of $o_{i+1}$ hold their tops and are blockers of $o_i$ (indices modulo $k$), with leftover goods that are distinct within each tree: one of its own per leaf, or the goods of a common pool of $k$, the leaves of each tree taking them in turn. Every cycle of want and pair arrows there has exactly $k$ pair arrows, and no free agent can finish if and only if $k \leq 2^d$, the number of leaves of a tree. For $k = 2$ and $d = 1$ this is the instance of Section B. We omit the proofs of these two facts; both are machine-checked in Lean for every $k$ and $d$.

## D Limits of the Shape

Could every instance have an $\mathrm{EFX}_0$ allocation with all bundles of at most two goods, or with one bundle of exactly three? No. Write $\theta_i(B)$, the *threat* of $B$ to $i$, for the largest value of $v_i(B \setminus \{h\})$ over $h \in B$ (0 if $B = \emptyset$): agent $i$ does not envy $X_j$ in the sense of $\mathrm{EFX}_0$ if and only if $v_i(X_i) \geq \theta_i(X_j)$, and $i$ is *safe* if this holds for every $j \neq i$. By additivity, $\theta_i(B) = 0$ if $|B| \leq 1$; $\theta_i(B) \leq v_i(B \cap R_i)$, with equality if $B$ has a good outside $R_i$; and $\theta_i(\{g, h\}) = \max(v_i(g), v_i(h))$.

**Lemma 8 (cases of safety).** *Let agent $i$ have $R_i = \{a, b, c\}$ with $v_i(a) > v_i(b) > v_i(c) > 0$ and $v_i(a) < v_i(b) + v_i(c)$, and let $X$ be an allocation. Then $i$ is safe in $X$ if and only if one of the following holds:*

**(T)** *$i$ holds $a$, and either $i$ also holds $b$ or $c$, or $b$ and $c$ are not together in a bundle of at least three goods;*
**(BC)** *$i$ holds $b$ and $c$;*
**(B)** *$i$ holds $b$, and $a$ is alone;*
**(C)** *$i$ holds $c$, and $a$ and $b$ are alone;*
**(E)** *$a$, $b$ and $c$ are all alone.*

*Proof.* Write $a$, $b$, $c$ also for the values. The sums of subsets of $\{a, b, c\}$ are ordered as $0 < c < b < a < b + c < a + c < a + b < a + b + c$. Let $S = X_i \cap R_i$.

- $S \supseteq \{a, b\}$ or $S \supseteq \{a, c\}$: every other bundle meets $R_i$ in at most one good, so its threat is at most $b < a + c$. Safe: (T).
- $S = \{b, c\}$: every threat is at most $a < b + c$. Safe: (BC).
- $S = \{a\}$: if $b$, $c$ are in different bundles, each threat is at most $b < a$; if $\{b, c\}$ is a bundle, its threat is $b < a$; if a bundle strictly contains $\{b, c\}$, it has a third good, outside $R_i$, and its threat is $b + c > a$. So: (T).
- $S = \{b\}$: if $a$ is not alone, its bundle has threat at least $a > b$; if $a$ is alone, every threat is at most $c < b$. So: (B).
- $S = \{c\}$: as before $a$ must be alone, and if $b$ is not alone its bundle has threat at least $b > c$. So: (C).
- $S = \emptyset$: a bundle with a good of $R_i$ and at least two goods has a positive threat. So: (E).

Conversely, each of (T), (BC), (B), (C), (E) describes safe situations above: (T) is the first and third items; (BC) is $S = \{b, c\}$ or $S = \{a, b, c\}$; (B) forces $a \notin S$, so $S = \{b\}$ with $a$ alone, or $S = \{b, c\}$; (C) forces $S = \{c\}$ with $a$ and $b$ alone; and (E) means that $S$ is empty or a single alone good, which is (T), (B) or (C). □

**Proposition 2 (limits of the shape).** *Take values with $v_i(a_i) > v_i(b_i) > v_i(c_i) > 0$ and $v_i(a_i) < v_i(b_i) + v_i(c_i)$ for every agent.*

*(a) Let agents 0, 1, 2 rank $g_0 \succ g_1 \succ p_i$, where $p_i$ is valued only by agent $i$. No EFX$_0$ allocation has all bundles of at most two goods, and $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$, to the three agents in any order, is EFX$_0$.*
*(b) Let agents 0, 1, 2, 3 rank $g_0 \succ g_2 \succ p_0$, $g_0 \succ g_2 \succ p_1$, $g_1 \succ g_2 \succ p_2$ and $g_1 \succ g_2 \succ p_3$, with $p_i$ valued only by agent $i$. Every EFX$_0$ allocation has bundle sizes $4, 1, 1, 1$, and one exists.*

*Proof.* (a) Five goods in three bundles of at most two goods have sizes $2, 2, 1$, so exactly one good is alone. Two agents do not hold their top $g_0$: case (T) fails for them, and (C) and (E) need two or three alone goods. So both are in case (BC) or (B), and both hold $g_1$, which is impossible. In $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$ the three agents are in cases (T), (B) and (C).

(b) Each of the tops $g_0$, $g_1$ has an agent ranking it first that does not hold it, a *loser*; the two losers differ. A loser is safe only by (BC) or (B), which need it to hold $g_2$, or by (C) or (E), which need $g_2$ alone. The two losers cannot both hold $g_2$, so $g_2$ is alone, and nobody is in case (BC). A loser holding $\{g_2\}$ needs its top alone; a loser not holding $g_2$ needs its top and $g_2$ alone. So $g_0$, $g_1$, $g_2$ are alone, held by three agents, and the fourth bundle is $\{p_0, p_1, p_2, p_3\}$. Such an allocation exists: $g_0$ to agent 0, $g_1$ to agent 2, $g_2$ to agent 1, the four goods $p_i$ to agent 3 (cases T, B, T, C). □

In (a) every agent values exactly three goods and is strictly balanced, so the one large bundle of Theorem 1 cannot be avoided; by (b), its size cannot be bounded by three.

## E Verification in Detail

*Lean.* The proofs of Sections 2–6 are machine-checked in core Lean 4 (`lean/EFX/K3DE*.lean` in [10], and `K3Extras.lean` for serial dictatorship): no library beyond Lean's own, no unproved step, every declaration on Lean's three standard axioms, checked by the repository's `lean/check.sh`, which also replays the library in a fresh kernel. DE is a Lean function, and the main theorem states that on every instance with natural-number values in which every agent positively values at most three goods it returns an $\mathrm{EFX}_0$ allocation in which all bundles but at most one have at most two goods, after at most $n$ peeling rounds and $4n$ trades. The Lean files use other names for some notions: *needs* for wants, *junk* for leftover goods, *utility* for score, *exposed* for blocker, *protecting good* for the leftover good $h_x$, *need arcs* and *exposure arcs* for want and pair arrows, *need cycle* and *exchange cycle* for rings, *pair chain* for a chain, (P) for (NC), and *valid absorber* for an agent that can finish (a notion that also allows a pair holder to finish, as when every agent holds its pair). One Lean lemma covers chains and rings (a chain is closed by an arrow from its last agent back to $x$).

Further files check the rest: serial dictatorship for at most two relevant goods; DE on ordered values such as the nonnegative reals (it runs on natural-number values computed from the given ones with at most

$n(m + 12)$ comparisons, which answer every comparison of two subset sums of an agent as the given values do, so its output is $\mathrm{EFX}_0$ for the given values); the facts of Appendix B and Figure 1 (by evaluation); an example of an allocation that is EFX but not $\mathrm{EFX}_0$; and the results of the appendices. The running time is checked on a counted program, proved to return exactly DE's output: at most $750(n+1)(n+m+1)$ units, reading the input included, with one unit per elementary operation, comparison, array read and array write, as in a random-access machine, arrays being used single-threaded. That the units match a machine is a property of the counted program, checked by reading it. The formal statements were written by AI agent sessions of the coding assistant that helped with this work (Claude Code) and compared with the text by independent AI sessions (`k3/simplify/po/referee/lean\_audit*.md` in [10]).

*Tests.* The tests are evidence, not proof. Three implementations, each asserting every claim of a proof and checking outputs against the raw definition of $\mathrm{EFX}_0$ with exact arithmetic, ran without failure on every profile with $n = 2$ and $m \leq 6$, with $n = 3$ and $m \leq 7$, and with $n = 4$ and $m \leq 6$ (in part), on cores with $n = 5, 6$ and on random profiles with up to 9 agents: about 1.7 million valid states for the proof of Section 5, about 15.5 million for an independently found second proof (a walk instead of a choice of distinct leftover goods), and about 2.9 million runs of a referee's implementation written from the text alone (`k3/simplify/po/` in [10]). The first and the referee's run DE's loop with the same three steps (up to the choice among leftover goods); the second tests the lemmas of Sections 4 and 5 on its own loop, which lets a free agent holding nothing finish first. The script `paper/k3-simple/examples/check\_examples.py`, written from this paper with the loop as stated here, recomputes every example and number of the paper and runs DE, peeling included, on random instances with ties, unbalanced agents and goods nobody values.

## F Related Work in a Table

**Table 1.** Existence results for complete allocations under restrictions. *Version cited*: the published version where one exists, else arXiv. [a]Goods nobody values go to an arbitrary agent. [b]Includes additive. [c]Extended abstract with a proof sketch. [d]An AAMAS 2025 extended abstract has the degree and girth conditions. [e]Accepted at EC 2026.

| Result | Restriction (agents, goods, who values what) | Notion | Complete | Valuations | Algorithm | Version cited |
|---|---|---|---|---|---|---|
| [5] | $n = 3$ | $\text{EFX}_0$ | yes | additive | pseudo-polynomial | JACM 2024 |
| [7] | at most three distinct valuations | $\text{EFX}_0$ | yes | additive | no bound stated | EC 2025 |
| [12] | $m \leq n + 3$ | $\text{EFX}_0$ | yes | monotone | none stated | MOR 2024 |
| [2] | $n = 4$, $m \leq 9$ | $\text{EFX}_0$ | yes | additive | none stated | arXiv |
| [4] | $m = n + 4$ | $\text{EFX}_0$ | yes, claimed | additive | none stated | Zenodo, not peer-reviewed |
| [6] | graphs: at most two valuers per good, two agents share at most one good | $\text{EFX}_0$ | yes | monotone | polynomial time | MOR 2026 (EC 2023) |
| [1] | multigraphs: at most two valuers per good | $\text{EFX}_0$ | yes | cancelable[b] | polynomial time | arXiv[e] |
| [13] | multigraphs: bipartite, or at most $\lceil n/4 \rceil - 1$ neighbours per agent, or girth at least six (non-parallel edges) | $\text{EFX}_0$ | yes | monotone | not efficient (stated) | arXiv[d] |
| [8] | at most two valuers per good, $v_i(g) \in \{0, v_g\}$ | $\text{EFX}_0$ | yes | restricted additive | no time bound stated | WINE 2024 |
| [9] | hypergraphs of girth at least four (a good joins its valuers) | $\text{EFX}_0$ | yes | monotone | polynomial time | AAAI 2026 |
| [14] | at most four relevant goods per agent | EFX | yes[a] | additive | polynomial time, announced | AAMAS 2024[c] |
| [11] | at most two relevant goods per agent | $\text{EFX}_0$ | yes | additive; monotone | serial dictatorship; Lean-checked | GitHub, v1.2.0 |
| this paper | at most three relevant goods per agent; any number of valuers per good | $\text{EFX}_0$ | yes | additive | polynomial time, at most $4n$ trades | this paper |